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DA3411 Business Valuation and Analysis

Individual Assignment 01

“The Fall of Lehman Brothers”

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The Fall of Lehman Brothers

1. Introduction

What is Lehman Brothers?

- Lehman Brothers was a global financial services firm, founded in 1850.
- It operated in investment banking, equity and fixed-income sales/trading, research, asset management, and private equity.
- By 2007, it was the 4th-largest investment bank in the U.S., with over \$600 billion in assets.

Lehman had grown aggressively by leveraging its capital and expanding into mortgage-backed securities and real estate investments.

What Happened in 2008?

In September 2008, Lehman Brothers filed for bankruptcy, the largest in U.S. history.

The collapse triggered a global financial panic, freezing credit markets and accelerating the Great Recession.

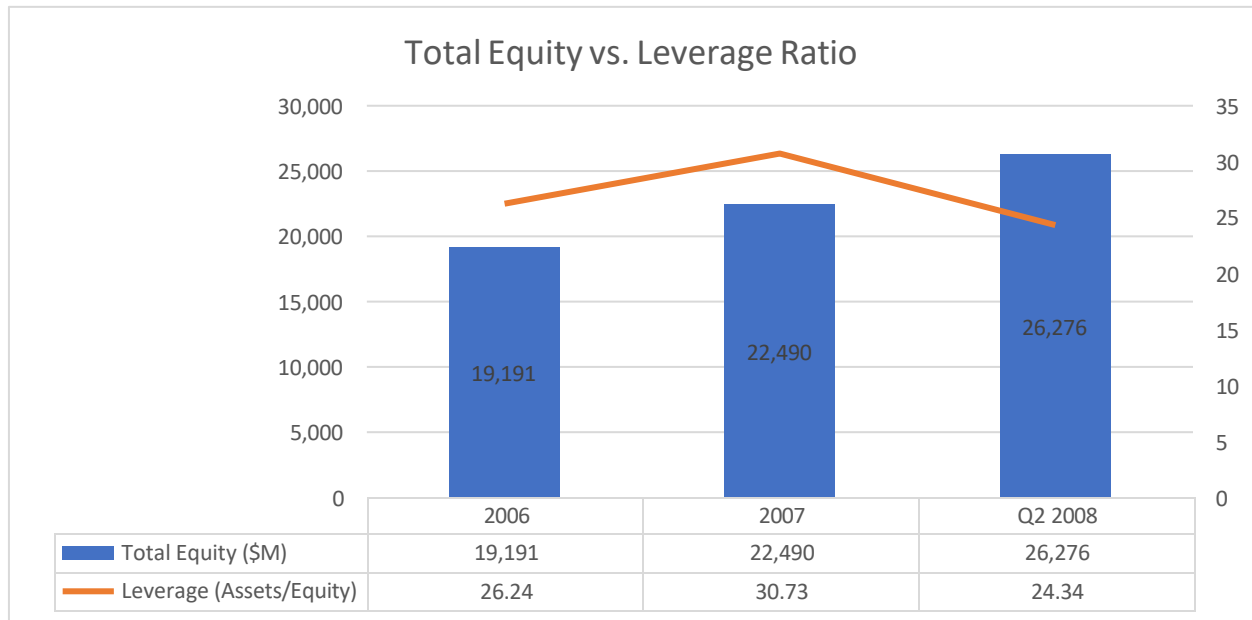
Unlike Bear Stearns and Merrill Lynch, Lehman received no government bailout.

The firm's failure stemmed from:

- Excessive leverage (over 30×)
- Overexposure to subprime mortgage assets
- Illiquidity during market stress
- Loss of investor and lender confidence

Lehman's fall symbolized the breaking point of the global financial system, leading to widespread reforms in banking and financial regulation.

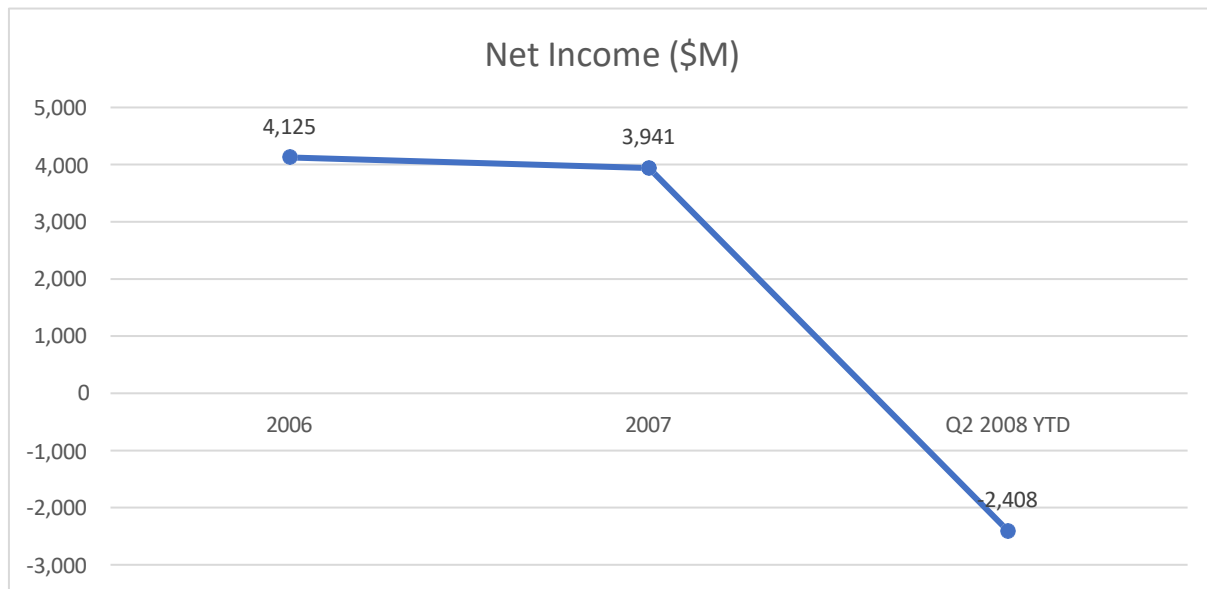
2. Financial Position (2006–2008)



Lehman Brothers' equity steadily increased from \$19.2B in 2006 to \$26.3B by mid-2008. However, this was outpaced by a dangerously high leverage ratio, peaking at 30.73 in 2007.

Although leverage fell in 2008 to 24.34, this was not due to risk reduction, but rather a drop in total assets and mounting losses. The firm remained heavily reliant on borrowed funds, leaving it extremely vulnerable to liquidity shocks.

3. Profitability Analysis



Between 2006 and 2007, Lehman Brothers maintained strong profitability, reporting net income of \$4.1 billion in 2006 and \$3.9 billion in 2007. This performance was driven by rapid expansion in structured finance products, particularly mortgage-backed securities (MBS) and real estate-related investments, which yielded high returns during the credit boom.

However, this profitability masked critical underlying weaknesses in its business model:

- While competitors began recognizing subprime-related losses in late 2007, Lehman deferred similar write-downs, preserving its earnings at the cost of transparency.
- The market began to doubt the reliability of its reported profits, especially as asset valuations became more opaque.
- Simultaneously, rising funding costs and worsening credit conditions eroded margins and investor trust

By Q2 2008, Lehman reported a net loss of \$2.4 billion, driven by:

- Significant impairments on MBS and commercial real estate holdings,
- Declining revenue in investment banking and trading,
- Liquidity pressures caused by repo market contraction and margin calls.

Profitability Ratios

Ratio	Calculation 2007	Result
Return on Equity(ROE)	Net Income/ Equity (3.94 B/22.49B)	17.50%
Net Profit Margin	Net income/Revenue (3.94B / 59B)	6.7%

Return on Equity: A solid ROE in 2007 indicated strong returns on capital but it was driven by unsustainable, highly leveraged returns from risky assets

Net Profit Margin: A 6–7% margin appears healthy, but it was largely built on earnings from overvalued structured products, which quickly unraveled.

The shift from strong profitability in 2006–2007 to severe losses in 2008 reflects how Lehman's earnings were overstated by risky, mark-to-market-sensitive assets. These profits were not resilient; they collapsed as asset prices fell and market liquidity dried up.

4. Asset Quality & Risk Exposure

From 2006 to mid-2008, Lehman Brothers significantly increased its exposure to risky and illiquid asset classes, notably mortgage-backed securities (MBS), real estate holdings, and derivative instruments. This shift in portfolio composition heightened the firm's vulnerability to market downturns and liquidity constraints.

Observations:

Mortgage-Backed Securities (MBS) and ABS Exposure

- Increased from \$57.7 billion in 2006 to \$89.1 billion in 2007 , a 55% year-over-year rise
- Slight reduction to \$72.5 billion by Q2 2008, but still represented a substantial concentration amid the collapsing U.S. housing market

Real Estate Held for Sale

- Rose sharply from \$9.4 billion in 2006 to \$21.9 billion in 2007, indicating aggressive expansion into direct property assets.
- Occurred during a period of rising mortgage defaults, further exposing the firm to potential write-downs

Derivatives Exposure

- Doubled from \$22.7 billion in 2006 to \$46.99 billion by Q2 2008.
- These instruments were complex and difficult to value, increasing the risk of mispricing and unforeseen counterparty losses.

Decline in Low-Risk, Liquid Assets

- Holdings in government and agency securities fell from \$47.3 billion in 2006 to \$27.0 billion in Q2 2008, reflecting a strategic shift away from safer assets.

Ratios & Risk Indicators:

Ratio	Calculation (2007)	Result
Risk Asset Ratio	(MBS + Real Estate + Derivatives) / Total Assets (\$155.6B / \$700B)	22.20%

Derivatives-to-Equity Ratio	\$44.6B / \$22.5B	1.98
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Risk Asset Ratio: Indicates that over one-fifth of Lehman's asset base consisted of volatile and illiquid instruments, an alarmingly high level for a firm dependent on daily funding.

Derivatives-to-Equity Ratio: For every dollar of equity, Lehman carried nearly \$2 of derivative risk, creating a significant threat to solvency in the event of revaluation losses.

By mid-2008, Lehman's asset composition was heavily skewed toward high-risk, low-liquidity instruments. This left the firm unable to respond effectively to market shocks or funding withdrawals. The interplay between asset illiquidity, valuation volatility, and funding fragility played a central role in accelerating Lehman's collapse.

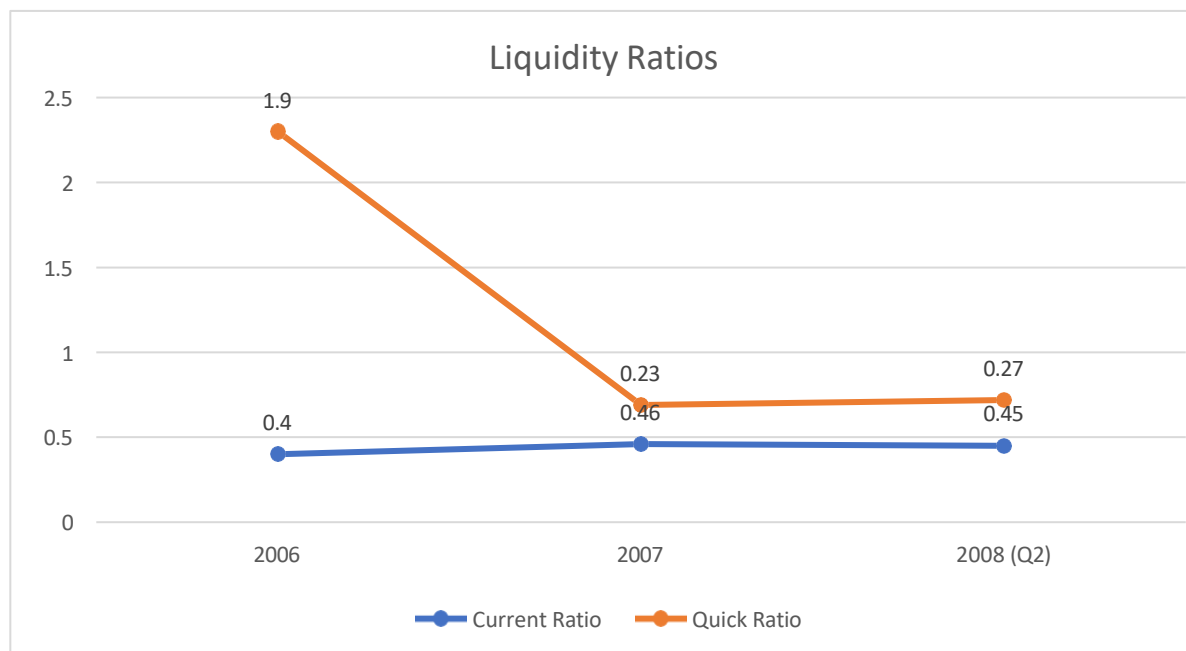
5. Liquidity & Solvency Analysis

From 2006 to 2008, Lehman Brothers' liquidity and solvency indicators signaled increasing financial vulnerability, even as the firm reported growing equity and business expansion. A detailed analysis of liquidity and debt metrics reveals a clear trend of deteriorating financial flexibility, particularly under stress conditions.

Liquidity Ratios

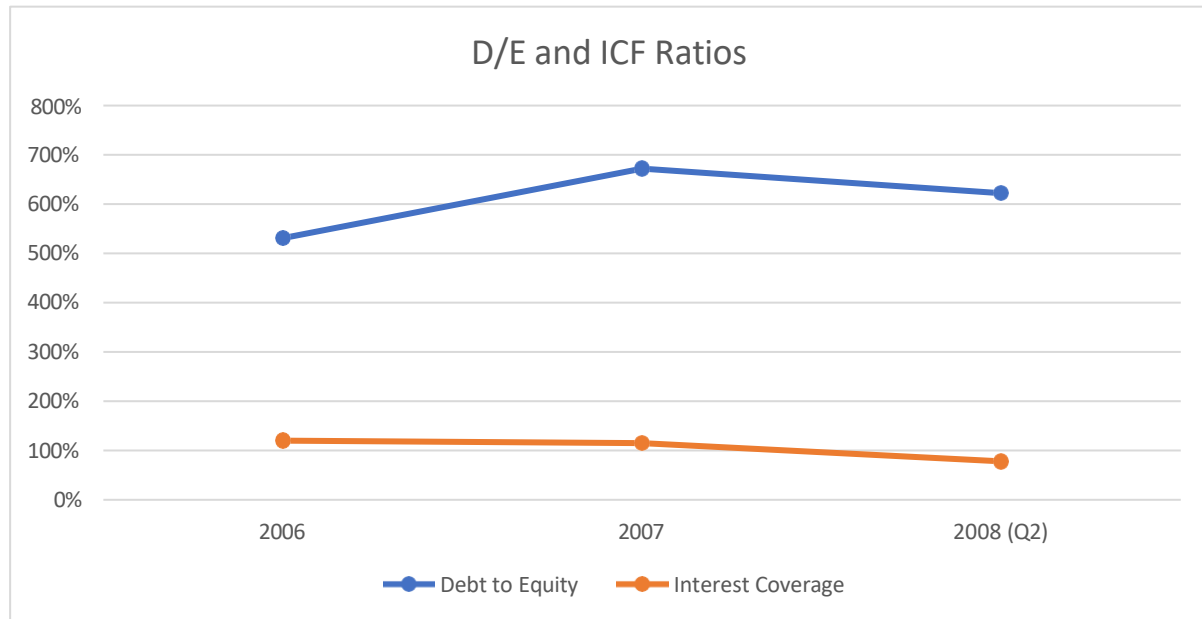
Ratio / Metric	2006	2007	2008 (Q2)
Current Ratio	0.4	0.46	0.45
Quick Ratio	0.19	0.23	0.27

Cash & Equivalents (\$M)	5,987	7,286	6,513
Current Assets (\$M)	40,049	63,306	61,265
Current Liabilities (\$M)	100,659	137,775	135,545
(Current Assets – Illiquid)	19,527	31,034	36,245



Solvency Ratios

Ratio / Metric	2006	2007	2008 (Q2)
Debt to Equity	531%	672%	622%
Debt to Capital	84%	87%	86%
Interest Coverage	1.2	1.15	0.78



From 2006 to mid-2008, Lehman Brothers operated with consistently weak liquidity and declining solvency strength, despite growing equity on paper. The firm's current ratio remained below 0.5, meaning it lacked the liquid current assets required to pay its short-term liabilities a major red flag for creditors and investors

Lehman's quick ratio plunged from 1.90 to 0.23 between 2006 and 2007, demonstrating how illiquid many of its "current" assets became as the credit markets tightened.

Solvency metrics also:

- The debt-to-equity ratio peaked at 672%, showing an unsustainable reliance on borrowed funds.
- The interest coverage ratio fell below 1.0 in 2008, indicating insufficient earnings to service debt, even before taxes and asset impairments.

With a balance sheet highly dependent on short-term repo agreements, Lehman was unable to rollover debt or raise emergency liquidity. This set the stage for a rapid collapse in solvency as counterparties and investors lost confidence.

High leverage, minimal liquid assets, and an inability to cover interest costs left the firm incapable of surviving even modest market stress. When short-term lenders pulled back, Lehman lacked both the liquidity and capital buffer to respond directly triggering its failure.

6. Timeline of Collapse, Key Events Leading to Lehman's Bankruptcy

The collapse of Lehman Brothers was not the result of a single event, but a series of missteps, external pressures, and delayed actions that unfolded over several months. The timeline below highlights key milestones that led to the firm's historic failure.

1. Q4 2007: Early Warning Signs Ignored

- While other major institutions began recognizing losses related to subprime mortgages, Lehman Brothers delayed writing down its risky assets.
- This created growing skepticism among investors and analysts regarding the accuracy of Lehman's financial disclosures

2. March 2008: Bear Stearns Collapse

- Bear Stearns was rescued by JP Morgan with federal backing.
- This event shifted market attention toward Lehman Brothers as the next likely casualty in the unfolding credit crisis

3. June 2008: First Major Loss

- Lehman reported a \$2.4 billion quarterly loss, its first in over a decade

- Management announced a \$6 billion capital raise, but the move failed to restore investor confidence

4. September 10, 2008: Strategic Restructuring Plan

- Lehman unveiled a last-minute restructuring strategy, including the spin-off of its real estate assets and partial sale of its asset management
- The market responded negatively, viewing the plan as too little, too late

5. September 12–14, 2008: Failed Rescue Talks

- Emergency meetings were held with government regulators and potential buyers (Barclays, Bank of America).
- Both buyers withdrew when the U.S. government refused to provide a financial backstop, similar to the one offered to Bear Stearns

6. September 15, 2008: Bankruptcy Filing

- Lehman Brothers filed for Chapter 11 bankruptcy protection, listing over \$600 billion in liabilities
- It became the largest bankruptcy in U.S. history, triggering global financial turmoil and a widespread credit freeze

Lehman Brothers' downfall was the result of compounded strategic misjudgments, declining market trust, and a lack of emergency support.

As liquidity dried up and losses mounted, the firm was unable to secure a buyer or bailout in time, resulting in a catastrophic failure that shook the global financial system.

7. Reasons Behind the Bankruptcy

Excessive Leverage

- Lehman operated with a leverage ratio exceeding 30:1, meaning even a 3% decline in asset value could wipe out its equity.
- This left the firm extremely vulnerable to asset price volatility.

Overexposure to High-Risk Assets

- Lehman invested heavily in subprime mortgage-backed securities, real estate, and complex derivatives.
- These assets became illiquid and massively devalued as the housing crisis unfolded.

Inadequate Liquidity Management

- Despite a balance sheet of over \$600 billion, Lehman had only about \$11 billion in accessible liquidity.
- When confidence eroded, the firm could not meet short-term obligations or roll over its debt.

Delayed Recognition of Losses

- Lehman postponed writing down toxic assets, creating mistrust among counterparties and investors.
- This hesitation damaged its credibility and made it harder to raise capital when it was most needed.

Lack of a Government Backstop

- Unlike Bear Stearns and AIG, Lehman was not offered a federal guarantee or rescue package.
- Without a buyer or bailout, bankruptcy became the only option.

Global Impact

- **Market Panic:** Lehman's failure triggered a worldwide selloff, froze credit markets, and led to a massive decline in global stock indices.
- **Financial Contagion:** Other banks and institutions faced similar liquidity crises, requiring unprecedented government interventions.
- **Regulatory Reforms:**
 - Introduction of Dodd-Frank Act (2010) in the U.S.
 - Creation of "too big to fail" resolution frameworks and stricter capital requirements globally.
- **Investor Behavior Shift:** Greater emphasis on liquidity, risk transparency, and stress testing in modern banking.

Lehman Brothers collapsed because of a toxic combination of excessive leverage, risk concentration, poor transparency, and inadequate liquidity. Its failure highlighted deep weaknesses in the global financial system, prompting a new era of regulatory overhaul, risk awareness, and institutional accountability.

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